

SAKET VARANASI

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EDUCATION

University of Pittsburgh — B.S. Electrical Engineering

Graduating Spring 2026

Dean's List (Fall 2024, Spring 2025) • Coursework: Analog & RF Circuits, Embedded Systems, Digital Logic & VHDL, Circuit Analysis, Signals & Systems, Machine Learning

TECHNICAL SKILLS

Sensor Characterization & Calibration: Independent sensor characterization experiments (half-bridge ADC measurement circuits, controlled gas delivery via syringe injection, full response curve and threshold mapping), multi-sensor PCB integration (temperature, humidity, pressure, light, pH, gas, RF power), sensor calibration and bias investigation, ADC-based signal conditioning, analog signal chain validation

Environmental Test & Data Analysis: Environmental chambers (temperature cycling — characterized capacitor behavior changes at low temperatures on RF energy-harvesting boards), Python (data analysis, statistical analysis, automated test scripts, large dataset processing, CSV logging, scripting), Linux/Bash (daily terminal use for embedded system integration, script management, codebase consolidation across multiple contributors), Git

Hardware & Fabrication: SolidWorks (3D enclosure and jig design — including flexible TPU/NinjaFlex enclosures for gas-tight sensor test chambers), 3D printing (Prusa, Bambu Labs — functional prototypes and production tooling), Altium Designer (schematic capture, PCB layout, custom library components), JLCPCB fabrication and assembly, SMD hand soldering and PCBA rework (0201/0402 components, trace cutting, jumper wires)

Embedded & Linux: Embedded Linux (NVIDIA Jetson — daily Bash scripting to integrate perception pipeline and collaborator code into unified main script, conda environment management, terminal-based debugging), ROS 2 (pub/sub nodes, service interfaces, cross-subsystem integration), Python, Git (version control across remote/local development), ESP32, Raspberry Pi, Arduino, STM32, I²C/SPI/UART/CAN bus

Documentation & Collaboration: Sensor characterization methodology documentation, FDA compliance documentation, test procedure development, revision-controlled design files, design review presentations, MS Office (Excel, Word, PowerPoint), SolidWorks, cross-functional team coordination

ENGINEERING EXPERIENCE

Pitt Idea Lab — Robotics Research — Computer Vision Engineer

Jan 2026 – Present

- Use Linux and Bash scripting daily to integrate the pothole detection perception pipeline with collaborators' code into a unified main script — manage conda environment on an NVIDIA Jetson, consolidate contributions from multiple developers into a clean executable pipeline, and debug integration issues from the terminal. Manage collaborative Git codebase with version control across local laptop and Jetson.
- Built end-to-end Python computer vision pipeline (Open3D, OpenCV, ROS 2) on a physical autonomous robot running embedded Linux — interfaced Basler ToF camera via PyPylon, applied RANSAC ground plane fitting, OpenCV contour detection, volumetric estimation. Validated across 20+ real-world field test cases.

Powercast Corporation — RF Electrical Engineering Intern

Jan 2025 – Sep 2025

- Ran environmental chamber testing for IC qualification — temperature-cycled RF energy-harvesting boards across operating ranges and characterized capacitor behavior changes at low temperatures. Observed and documented how storage capacitor charge/discharge behavior shifted in cold conditions, affecting the timing of the charge-and-fire startup sequence. This directly informed design decisions about component selection and threshold settings.
- Designed 915 MHz RF energy-harvesting PCB in Altium, performed extensive SMD hand soldering and PCBA rework (0201/0402 components), and root-caused a DC power-path reliability failure. Designed load switch PCB fix with hysteretic UVLO energy gating, validated end-to-end, and presented corrective action in formal design review. Achieved 33% sensitivity improvement over baseline.
- Built Python automated test system for temperature cycling, voltage sweep, and yield characterization — automated pass/fail against production spec limits, reduced test turnaround 30%. Maintained FDA-compliant documentation throughout a 9-month product cycle.

BPMI — Environmental Monitoring — Sensor Design Engineer

Jun 2024 – Dec 2024

- Designed multi-sensor PCB integrating five sensor types (pH, temperature, humidity, moisture, light) with ESP32 wireless transmission — selected sensor suite based on independent field research, sized I²C data bus to accommodate evolving requirements, calibrated pH sensor with buffer solutions, validated full sensor signal chain before client delivery.
- Designed complete solar-powered power system from scratch — CN3791 MPPT charger, Li-ion cell with DW01A BMS, 5V-to-3.3V step-down regulator. Chose MPPT over linear charging based on efficiency analysis at 1W panel output.

Panther Racing — Formula SAE — Electronics Design Engineer & Harness Lead

Jan 2024 – Present

- Designed SolidWorks jigs and 3D printed functional electronics enclosures for the race vehicle — iterated on mechanical fit and tolerances across multiple fabrication cycles. Introduced RapidHarness for wire-level traceability, developed manufacturing standards from scratch, validated full harness before power-on. Zero electrical failures at competition.
- Designed CAN-bus steering-wheel PCB in Altium end-to-end, defined full vehicle PDN, designed SolidWorks jigs and 3D printed enclosures. Coordinated daily across mechanical, drivetrain, cooling, and controls sub-teams.

PROJECTS

Sensor Project (Li-ion Tamer / Honeywell) — Li-ion Thermal Runaway Gas Sensor — Self-Initiated R&D 2024

- Cold-contacted Li-ion Tamer (Honeywell) to perform independent sensor characterization — designed sealed NinjaFlex 3D-printed test enclosure for gas-tight sensor isolation, built half-bridge ADC measurement circuit to convert resistive sensor output to voltage readable by Arduino ADC, introduced gas via controlled syringe injection at incremental concentrations, characterized full voltage response curve and detection threshold, delivered documentation to the company. Designed the flexible enclosure specifically so it could self-seal.

Personal Project — Guitar Looper Pedal

Aug – Dec 2024

- Designed custom PCB and SolidWorks enclosure for a Raspberry Pi-based real-time audio looper — SMD hand-assembled, validated, and field-tested in live performance.

LEADERSHIP & EXTRACURRICULARS

Pitt Tennis — Team Captain • Food Drive Organizer, GPA Food Bank • Sports Photography • Guitar